

MAURITIUS VACOAS, MauritiusWMO: **619950**Lat: **20.2971S**Lon: **57.4969E**Elev: **425**StdP: **96.32**Time Zone: **4.00 (E04)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.0	15.6	11.4	8.8	17.6	12.3	9.4	17.9	8.6	19.5	8.0	19.4	2.3	130	0.298

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.9	28.7	23.7	28.1	23.4	27.6	23.1	24.8	26.9	24.4	26.6	24.1	26.3	3.2	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.2	20.1	25.9	23.8	19.6	25.5	23.4	19.2	25.3	78.0	27.2	76.3	26.7	74.9	26.2	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	7.1	6.4	12.8	30.2	1.1	0.8	12.0	30.8	11.4	31.2	10.8	31.6	10.0	32.2		
(5)			11.7	25.8	1.5	0.7	10.7	26.4	9.8	26.8	9.0	27.2	8.0	27.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.0	24.6	24.8	24.4	23.6	21.6	20.0	19.1	19.1	19.6	20.9	22.3	23.8		
	DBStd	2.40	0.90	0.95	0.96	1.00	1.28	1.26	1.15	1.10	1.12	1.16	1.22	1.00		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	14	0	0	0	0	0	2	4	4	3	1	0	0		
	CDD10.0	4371	453	414	447	407	359	299	283	282	288	337	371	429		
	CDD18.3	1342	195	181	189	157	101	51	29	28	41	79	121	171		
	CDH23.3	4533	931	893	831	491	158	28	9	7	16	96	341	732		
	CDH26.7	381	106	103	77	19	3	1	1	0	0	1	8	62		
(14) Wind	WSAvg	3.4	3.4	3.2	3.2	3.3	3.1	3.5	3.9	3.9	3.6	3.5	3.1	3.1		
(15) Precipitation	PrecAvg	1448	213	246	219	137	111	72	89	66	58	43	64	130		
	PrecMax	2263	644	641	651	352	300	175	156	112	248	207	318	535		
	PrecMin	742	0	55	5	0	23	20	25	4	2	7	10	0		
	PrecStd	303	159	155	141	81	69	38	33	32	48	42	60	128		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.2	29.4	29.0	28.1	27.1	25.4	24.4	24.1	24.8	26.2	27.7	28.9		
		MCWB	23.8	24.3	24.1	23.3	22.5	20.7	19.4	19.3	19.5	20.4	22.1	23.2		
	2%	DB	28.5	28.6	28.2	27.2	25.9	24.1	23.1	23.0	23.7	25.1	26.7	28.0		
		MCWB	23.5	24.0	23.6	22.8	21.7	19.9	19.0	18.6	18.9	20.1	21.5	22.7		
	5%	DB	27.8	27.9	27.6	26.6	25.0	23.2	22.2	22.1	22.9	24.4	26.1	27.4		
		MCWB	23.3	23.8	23.4	22.5	21.1	19.4	18.6	18.1	18.5	19.7	21.1	22.3		
	10%	DB	27.1	27.2	26.9	25.9	24.2	22.4	21.4	21.4	22.2	23.7	25.3	26.7		
		MCWB	23.0	23.6	23.1	22.2	20.7	18.9	18.2	17.8	18.2	19.3	20.7	22.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.0	25.5	25.2	24.3	23.8	21.7	20.7	20.5	20.6	22.0	23.2	24.4		
		MCDB	27.5	27.5	27.2	26.5	25.7	24.3	22.6	22.4	22.7	24.2	25.9	27.4		
	2%	WB	24.5	25.0	24.5	23.7	22.6	20.6	19.8	19.6	19.7	21.1	22.5	23.7		
		MCDB	26.9	27.1	26.7	25.8	24.7	22.7	21.8	21.6	22.2	23.6	25.2	26.6		
	5%	WB	24.1	24.6	24.1	23.2	21.9	20.1	19.2	19.0	19.2	20.5	21.9	23.2		
		MCDB	26.4	26.6	26.3	25.3	23.9	22.1	21.1	21.0	21.7	23.0	24.6	25.9		
	10%	WB	23.7	24.2	23.7	22.8	21.3	19.6	18.8	18.4	18.8	20.0	21.4	22.8		
		MCDB	25.9	26.1	25.8	24.8	23.3	21.5	20.6	20.4	21.2	22.5	24.0	25.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.1	4.9	5.0	4.9	5.2	5.0	4.8	4.9	5.4	5.7	6.0	5.9		
		MCDBR	6.1	5.9	5.9	5.7	6.3	6.1	6.0	5.9	6.3	6.5	6.8	6.6		
	5% WB	MCWBR	2.2	2.4	2.3	2.2	2.8	2.8	2.8	2.6	2.5	2.4	2.4	2.3		
		MCWBR	5.2	5.1	5.0	5.0	5.5	5.3	5.2	5.2	5.7	5.8	6.0	5.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.384	0.396	0.383	0.361	0.342	0.324	0.326	0.328	0.339	0.353	0.354	0.363		
		taud	2.559	2.521	2.558	2.608	2.641	2.648	2.643	2.625	2.560	2.536	2.573	2.595		
	Ebn at Noon	Ebn at Noon	961	937	926	908	889	889	898	929	953	968	983	981		
		Edh at Noon	109	112	104	92	82	79	81	89	101	109	107	105		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	6.76	6.34	5.88	5.14	4.33	3.95	4.10	4.84	5.81	6.59	7.14	7.21		
	RadStd	RadStd	0.45	0.38	0.34	0.27	0.18	0.12	0.14	0.13	0.20	0.19	0.24	0.38		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		+0.60	+0.65	+0.40	+0.27	N/A	N/A	N/A	N/A	+216	+216

Nomenclature: See separate page